

Problem

For a Y-connected load, the time-domain expressions for three line-to-neutral voltages at the terminals are:

$$v_{AN} = 120 \cos(\omega t + 32^\circ) \text{ V}$$

$$v_{BN} = 120 \cos(\omega t - 88^\circ) \text{ V}$$

$$v_{CN} = 120 \cos(\omega t + 152^\circ) \text{ V}$$

Write the time-domain expressions for the line-to-line voltages v_{AB} , v_{BC} , and v_{CA} .

Step-by-step solution

Step 1 of 3

Consider the following data.

Three line to neutral voltages are,

$$v_{an} = 120 \cos(\omega t + 32^\circ) \text{ V}$$

$$v_{bn} = 120 \cos(\omega t - 88^\circ) \text{ V}$$

$$v_{cn} = 120 \cos(\omega t + 152^\circ) \text{ V}$$

Step 2 of 3

By converting above time-domain expression into frequency domain

$$\mathbf{V}_{an} = 120 \angle 32^\circ \text{ V}$$

$$\mathbf{V}_{bn} = 120 \angle -88^\circ \text{ V}$$

$$\mathbf{V}_{cn} = 120 \angle 152^\circ \text{ V}$$

We know

$$\begin{aligned} \mathbf{V}_{AB} &= \mathbf{V}_{an} - \mathbf{V}_{bn} \\ &= 120 \angle 32^\circ - 120 \angle -88^\circ \\ &= 101.76 + j63.59 - 4.187 + j119.92 \\ &= 97.58 + j183.51 \end{aligned}$$

$$= 207.84 \angle 62^\circ \text{ V}$$

$$\begin{aligned} \mathbf{V}_{BC} &= \mathbf{V}_{bn} - \mathbf{V}_{cn} \\ &= 120 \angle -88^\circ - 120 \angle 152^\circ \\ &= 4.187 - j119.92 + 105.95 - j56.33 \\ &= 110.14 - j176.26 \end{aligned}$$

$$= 207.8 \angle -58^\circ \text{ V}$$

$$\begin{aligned} \mathbf{V}_{CA} &= \mathbf{V}_{cn} - \mathbf{V}_{an} \\ &= 120 \angle 152^\circ - 120 \angle 32^\circ \\ &= -105.95 + j56.33 - 101.76 - j63.59 \\ &= -207.71 - j7.253 \end{aligned}$$

$$= 207.8 \angle -178^\circ \text{ V}$$

Step 3 of 3

Thus time domain expression for above line voltages are

$$\begin{aligned} v_{AB} &= 207.84 \cos(\omega t + 62^\circ) \text{ V} \\ v_{BC} &= 207.8 \cos(\omega t - 58^\circ) \text{ V} \\ v_{CA} &= 207.8 \cos(\omega t - 178^\circ) \text{ V} \end{aligned}$$